

141 Series

**Electronic
Printing
Calculators**

**Operating
Instructions**

Unicom

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FOREWORD

UNICOM series 141 Electronic Printing Calculators are efficient, reliable, and easy to use. The highly versatile UNICOM

141 handles a wide range of applications at electronic speed, and provides a printed record with complete identification of each entry.

It includes such features as: high speed printer, 14 digit capacity, input buffer, sub total accumulator, main total accumulator, one memory, automatic constant calculation, automatic round-off, protective keyboard interlocks, and up to eight decimal places.

The simplicity of operation enables anyone to master the UNICOM 141 in a few minutes. To add, subtract, multiply, or divide, you simply enter the amounts on the keyboard, and depress the function keys (+, -, x, $\div$, $\sqrt{}$) in the same sequence as in manual calculations. Be sure calculator is properly grounded. Use an adapter plug if needed. Air vents at the back of the machine should not be covered while the calculator is on. UNICOM 141 should not be placed in intense direct sun light or near heating devices. When turned off, all figures are cleared, including those in memory. UNICOM 141 Printing calculators are guaranteed for one full year.

* Square root model only.



SPECIFICATIONS

Read Out	Printer Capacity	14 digits plus decimal point and symbols
Operation speed	Addition (Subtraction)	0.45 sec
	Multiplication	1.1 sec
	Division	1.2 sec
Capacity of Input Buffer		8 words
Main Element	MOS-LSI	
Operating Temperature Guaranteed	(+32°F) to (+104°F)	
Paper	Width 2-1/4", Diameter 2-3/4"	
Power Source	AC 115V ±10%	
Power Requirement	20 watt	
Dimension	8.3" (W) x 13.2" (D) x 5.1" (H)	
Weight	13 pounds	

UNICOM 141 contains 5 working registers and 1 memory.

5 working registers;

1 Entry register.

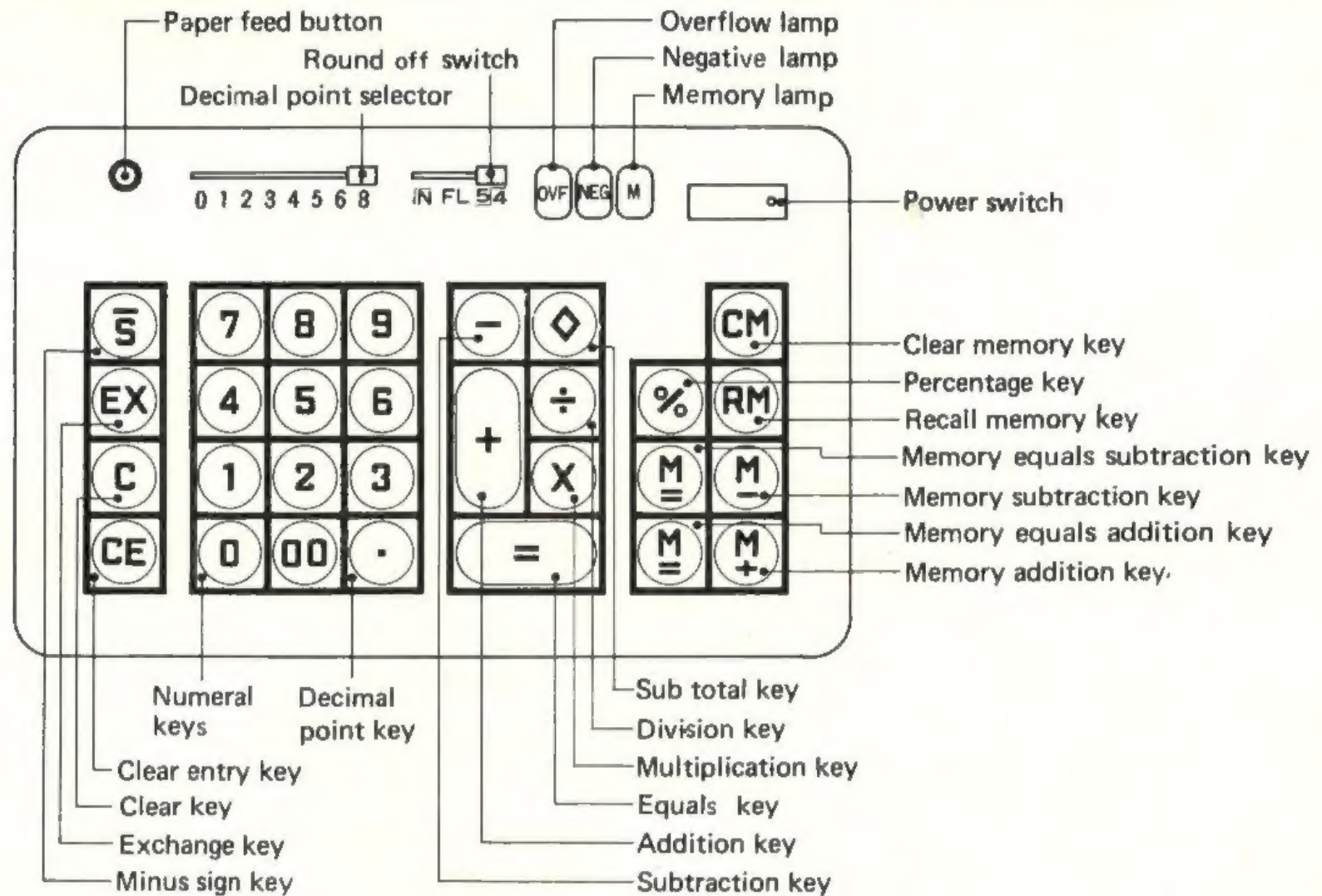
1 Sub total register.

1 Main total register.

2 Multiplication—division registers.

In multiplication (also division), the product (also quotient) is produced in the entry register and the both accumulators are unaffected.

KEYBOARD



OPERATING KEY FUNCTIONS

Key	Name	Explanation	Print Symbol
	Clear key	For clearing out working registers and overflow.	C
	Cleareentry key	For clearing out incorrect entry and overflow.	
~	Numeral keys		
	Decimal point key		
	Minus sign key	For entering negative factors.	
	Subtraction key	To subtract from sub/main total accumulators.	—
	Addition key	To add to sub/main total accumulators.	+
	Division key	To set divide mode, perform chain division and establish 2nd factor as constant divisor.	÷
	Multiplication key	To set multiply mode, perform chain multiplication and establish 1st factor as constant multiplicand.	×
	Equals key	For printing and clearing main total accumulator after the touch of addition or subtraction key.	=
		For calculating and printing product or quotient in multiplication or division.	✖

Key	Name	Explanation	Print Symbol
	Sub total key	For printing and clearing sub total accumulator after the touch of addition or subtraction key. For printing dates or reference numbers after the touch of numeral key. For printing intermediate results in chain calculations.	 # 
	Exchange key	For exchanging contents of multiplier and multiplicand (divisor and dividend)	Ex
	Percentage key	For percentage calculations.	% ✖
	Clear memory key	For printing and clearing memory.	CM
	Recall memory key	To recall and print (but not clear) memory.	RM
	Memory subtraction key	To subtract from memory.	<u>M</u>
	Memory addition key	To add to memory.	<u>M</u> +
	Memory equals subtraction key	For calculating and printing product or quotient and automatically subtracting from memory.	= <u>M</u>

Key	Name	Explanation	Print Symbol
[M]	Memory equals addition key	For calculating and printing product or quotient and automatically adding to memory.	= M +

Decimal point selector: For setting position of decimal point (0, 1, 2, 3, 4, 5, 6, 8)
 Round switch : For floating, rounding or truncating.
 Overflow lamp : Turns on when the results exceed the capacity.
 Negative lamp : Turns on when the entry or result is negative.
 Memory lamp : Turns on when amount is registered in memory.
 Paper feed button : For advancing the paper tape.
 Power switch : For power on/off.

CHANGING RECORDING PAPER ROLL

Appearance of a red paper section indicates the paper supply is running short. Standard tape 2-1/4" in width (2-3/4" in diameter) should be used as replacement.

Method of changing :

1. Remove the printing section cover by lifting the back of the cover. (Fig. 1)
2. Lift paper guide and feed paper tape between chrome plate and paper guide, depressing the paper feed button.
3. Insert paper into the paper guide slit, depressing the paper feed button. (Fig. 2)
4. Press down on paper guide until it clicks into place, tear off excess paper, and replace printing section cover. (Fig. 3)



Fig. 1



Fig. 2



Fig. 3

CHANGING PRINT RIBBON

The ribbon should be replaced after use of 5 ~ 6 rolls of tape. We recommend the following nylon ribbon.

Width 0,5" , length 24"

Changing the ribbon is performed as follows.

- 1) Remove the printing section cover by lifting the back of the cover. (Fig.1)
- 2) Press the check lever on the back side of each spool, and pull the spools upwards. (Fig.2)
- 3) Insert new ribbon supply and take-up spools with black half up, making sure both spools are snapped onto advance mechanism catches. Check levers should be held against ribbon by spring tension. (Fig. 3)
- 4) Replace the cover. (Fig. 4)



Fig. 1



Fig. 2



Fig. 3



Fig. 4

OPERATION EXAMPLES

1. ADDITION / SUBTRACTION

I I

$$12.34 - 34.56 + 56.789 + 56.789 + 0.123 \\ = 91.481$$

Determine the maximum number of decimals to be entered and set decimal point selector to that position.

To add, enter the amount on the keyboard and press the $\oplus$ key. The adjustment of the decimal point is automatically processed according to D.P. Selector. Addition and Subtraction are independent from Round off switch.

To subtract, enter the amount on the key board and press the $\ominus$ key.

※ To repeat, simply depress the previous key without re-entering it on the keyboard.

When the problem is completed, press the $\equiv$ key to print answer on the tape. Pressing $\equiv$ key clears the sub/main total accumulators.

I - I

Operation		Printed tape	
DP=3	$\overline{IN}$		
12.34	$\oplus$	12.340	+
34.56	$\ominus$	34.560	-
56.789	$\oplus$	56.789	+
※	$\oplus$	56.789	+
.123	$\oplus$	0.123	+
	$\equiv$	91.481	×

I-2

$$1.23 + 4.56 = 5.79$$

When necessary to obtain a result rounded to 1 decimal, at first, set the decimal point selector at more than 1 (for example DP=2). After the touch of last subtraction or addition key, re-set the decimal point selector to "1" and depress the $\square$ key.

※ When the result is rounded, the following symbol will be printed out: "1".

Credit Balance

I-3

$$15.3 - 56.789 + 3.456 = -38.033$$

Negative answers print in red.

I-2

I-3

Operation	Printed tape
DP=2 $\overline{54}$	
1.23 $\oplus$	1.23 +
4.56 $\oplus$	4.56 +
DP=1 $\overline{54}$	
※ $\square$	5.8 1 ※
DP=3 $\overline{N}$	
15.3 $\oplus$	15.300 +
56.789 $\ominus$	56.789 -
3.456 $\oplus$	3.456 +
$\square$	38.033 ※ red

Sub/main total accumulator and non-add printing

1-4 DP 2 $\overline{N}$

$$1+2+3=6$$

$$4+5+6=15$$

$$7+8+9=24$$

$$45$$

The UNICOM 141 contains two accumulators. The one is used as the sub total accumulator, the other is used as the main total accumulator.

When the problem is completed, press the $\square$ key to print the sub total accumulator on the tape. Pressing the $\diamond$ key clears out the sub total accumulator. The main total accumulator is unaffected.

※ Pressing the $\diamond$ key after the touch of numeral key prints the dates or reference numbers on the tape.

1-4

Operation		Printed tape	
DP=2 $\overline{N}$			
※	1 $\diamond$	1	#
	1 $+$	1.00	+
	2 $+$	2.00	+
	3 $+$	3.00	+
	$\diamond$	6.00	$\diamond$
※	2 $\diamond$	2	#
	4 $+$	4.00	+
	5 $+$	5.00	+
	6 $+$	6.00	+
	$\diamond$	15.00	$\diamond$
※	3 $\diamond$	3	#
	7 $+$	7.00	+
	8 $+$	8.00	+
	9 $+$	9.00	+
	$\diamond$	24.00	$\diamond$
	$\square$	45.00	※

2. MULTIPLICATION

2-1 FL

$$12.3 \times 4.56 = 56.088$$

2-2 DP=2 $\overline{54}$

$$12.3 \times 4.56 = 56.09$$

2-3 DP=2 $\overline{N}$

$$12.3 \times 4.56 = 56.08$$

D.P. of product (also quotient) is set at the place designated by D.P. Selector if Round Switch is $\overline{54}$ or $\overline{N}$, otherwise, D.P. is automatically set at the arithmetic position.

When the product quotient is rounded, the following symbol will be printed: "f".

Chain multiplication

2-4 FL

$$12.3 \times 4.56 \times 0.789 = 44.253432$$

In multiplication (also division), D.P. of intermediate product (also quotient) is automatically set when in float position regardless of the setting of D.P. selector. In continuous multiplication, simply enter the problem as it is written.

*The intermediate product (also quotient) can be printed on the tape by depressing the $\boxtimes$ key.

2-1

2-2

2-3

2-4

Operation	Printed tape
FL	
12.3 $\boxtimes$	12.3 $\times$
4.56 $\boxtimes$	4.56 $=$
	56.088 $\times$
DP=2 $\overline{54}$	
12.3 $\boxtimes$	12.3 $\times$
4.56 $\boxtimes$	4.56 $=$
	56.09 $\nearrow \times$
DP=2 $\overline{N}$	
12.3 $\boxtimes$	12.3 $\times$
4.56 $\boxtimes$	4.56 $=$
	56.08 $\times$
FL	
12.3 $\boxtimes$	12.3 $\times$
4.56 $\boxtimes$	4.56 $\times$
$\boxtimes$	56.088 $\diamond$
.789 $\boxtimes$	0.789 $=$
	44.253432 $\times$

Multiplication with constant multiplicand

2-5

FL

$$361.52 \times 120 = 43382.40$$

$$361.52 \times 118.6 = 42876.272$$

$$361.52 \times 98.4 = 35573.568$$

After completion of the first multiplication, the constant multiplicand (361.52) is retained in the multiplication-division register. Thereafter, simply enter the variable multipliers and depress the $\boxed{=}$ key.

Multiplication with constant multiplier

2-6

FL

$$1.25 \times 12 = 15.00$$

$$3.50 \times 12 = 42.00$$

$$1.99 \times 12 = 23.88$$

Before completion of the first multiplication, the multiplicand and multiplier must be exchanged by using the $\boxed{EX}$ key.

2 5

Operation	Printed tape
FL	
361.52 $\boxed{\times}$	361.52 $\times$
120 $\boxed{=}$	120 =
	43382.40 $\times$
118.6 $\boxed{=}$	118.6 =
	42876.272 $\times$
98.4 $\boxed{=}$	98.4 =
	35573.568 $\times$
FL	
1.25 $\boxed{\times}$	1.25 $\times$
12 $\boxed{EX}$	12 EX
$\boxed{=}$	1.25 =
	15.00 $\times$
3.50 $\boxed{=}$	3.50 =
	42.00 $\times$
1.99 $\boxed{=}$	1.99 =
	23.88 $\times$

Raising to power

2-7 FL

$$5^4 = 625$$

Following the entry, depress the $\times$ key. Then, depress the $=$ key once and the second power product is produced. The constant multiplicand (5) is still retained in the multiplication - division register. Thereafter, depress the $=$ key again and third power is produced, and so forth.

Correction of function order

2-8 FL

$$12.3 \times 4.56 = 56.088$$

After the touch of division (also multiplication) key, the mode of operation can be corrected by depressing the multiplication (also division) key.

2-7

2-8

Operation	Printed tape
FL	
5 $\times$	5 $\times$
$=$	5 $=$
	25 $\times$
$=$	25 $=$
	125 $\times$
$=$	125 $=$
	625 $\times$
FL	
12.3 $\div$	12.3 $\div$
$\times$	$\times$
$\div$	$\div$
$\times$	$\times$
4.56 $=$	4.56 $=$
	56.088 $\times$

3. DIVISION

3-1 Division with full floating decimal quotient.

3-2 Division with quotient rounded off.

3-3 Division with quotient truncated.

Chain division

3-4 FL

$$123 \div 6 \div 0.789 = 25.98225602027$$

In continuous division, simply enter the problem the same way it is written.

3-1

3-2

3-3

3-4

Operation	Printed tape
FL 40 $\div$ 6 $=$	40 $\div$ = 6.66666666666666 $\times$
DP=2 $\overline{54}$ 40 $\div$ 6 $=$	40 $\div$ 6 $=$ 6.67 $\nearrow \times$
DP=2 $\overline{N}$ 40 $\div$ 6 $=$	40 $\div$ 6 $=$ 6.66 $\times$
FL 123 $\div$ 6 $\div$.789 $=$	123 $\div$ 6 $\div$ 0.789 $=$ 25.98225602027 $\times$

Division by constant divisor

3-5 DP=2 54

$$4578 \div 360 = 12.72$$

$$2902 \div 360 = 8.06$$

$$8716 \div 360 = 24.21$$

After completion of the first division, the divisor (360) is retained in the multiplication - division register. Thereafter, simply enter the variable dividend and depress the $\boxed{=}$ key.

Division by constant dividend

3-6 DP=2 54

$$123.45 \div 36.9 = 3.35$$

$$123.45 \div 28.4 = 4.35$$

$$123.45 \div 31.55 = 3.91$$

The constant is retained in the memory and is recalled as a dividend by pressing the $\boxed{RM}$ and $\boxed{\div}$ keys as indicated.

Note: The memory indicator lights up when there are figures in the memory. To clear the memory, press the $\boxed{CM}$ key.

3-5

3-6

Operation	Printed tape
DP=2 54	
4578 $\boxed{\div}$	4578 $\div$
360 $\boxed{=}$	360 =
	12.72 $\nearrow \times$
2902 $\boxed{=}$	2902 =
	8.06 $\times$
8716 $\boxed{=}$	8716 =
	24.21 $\times$
DP=2 54	
123.45 $\boxed{M}$	123.45
$\boxed{\div}$	123.45 $\div$
36.9 $\boxed{=}$	36.9 =
	3.35 $\nearrow \times$
$\boxed{RM}$	123.45 RM
$\boxed{\div}$	123.45 $\div$
28.4 $\boxed{=}$	28.4 =
	4.35 $\nearrow \times$
$\boxed{RM}$	123.45 RM
$\boxed{\div}$	123.45 $\div$
31.55 $\boxed{=}$	31.55 =
	3.91 $\times$

4. PERCENTAGE CALCULATION

Percentage multiplication

4-1 FL

$$12345 \times 2 (\%) = 246.90$$

After completion of the multiplication, D.P. in product is placed as if .02 (2%) had been entered. The $\left(\frac{\%}{\square}\right)$ key speeds up entry of percentage factors.

4-1

Operation	Printed tape
FL	
12345 $\left(\frac{\%}{\square}\right)$	12345 $\times$
2 $\left(\frac{\%}{\square}\right)$	2 $\%$
	246.90 $\times$
2 $\left(\frac{\%}{\square}\right)$	2 $\div$
3 $\left(\frac{\%}{\square}\right)$	3 $\%$
	66.6666666666 $\times$

Percentage division

4-2 FL

$$2 \div 3 = 0.666666666666$$



$$66.6666666666 \%$$

After completion of the division, D.P. in quotient is placed to be read as a percentage rather than a decimal.

4-2

5. MIXED CALCULATION

5-1

$$\frac{(1.5 + 129.05 - 11.08) \times 12.4 \div 0.55}{(12.96 - 3.56) \times 0.87} = 329.36 \quad 5-1$$

After completion of addition or subtraction, a touch of the $\boxtimes$ (also $\div$) key reads and clears the main total accumulator, at the same time, orders multiplication or division.

Operation	Printed tape
DP=2 $\overline{54}$	
C	O C
1.5 $\oplus$	1.50 +
129.05 $\oplus$	129.05 +
11.08 $\ominus$	11.08 -
* $\boxtimes$	119.47 $\times$
12.4 $\div$	12.4 $\div$
.55 $\div$	0.55 $\div$
12.96 $\oplus$	12.96 +
** 3.56 $\ominus$	3.56 -
C	9.40 $\times$
$\div$	9.40 $\div$
0.87 C	0.87 =
	329.36 $\times$

6. PERCENTAGE DISTRIBUTION

6-1 DP=2 54

123	=	8.99 %
456	=	33.33 %
789	=	57.68 %
1368	=	100.00 %

※ A 100% proof may be obtained by adding individual percentage distributions into the accumulator without use of memory.

6-1

Operation	Printed tape
DP=2 <u>54</u>	
C	0 C
123 +	123 $\div$
123 +	123.00 $+$
456 +	456.00 $+$
789 +	789.00 $+$
=	1368.00 $\times$
$\text{\%}$	1368.00 $\%$
	8.99 $\times$
※ +	8.99 $+$
456 $\text{\%}$	456 $\%$
	33.33 $\times$
※ +	33.33 $+$
789 $\text{\%}$	789 $\%$
	57.68 $\times$
※ +	57.68 $+$
※ =	100.00 $\times$

REVERSED CALCULATION

6-2

3
(1.23×4) + (5.67×8) =0.05

Individual products may be accumulated to a grand total in the accumulator without use of memory.

6 2

Operation	Printed tape
DP=2	
	0 C
1.23	1.23 X
4	4
	4.92 X
	4.92 +
5.67	5.67 X
8	8 =
	45.36 X
	45.36 +
	50.28 .
3	3 Ex
	50.28 —
	0.05 X

7. MULTIPLICATION BY CONSTANT WITH ACCUMULATION

7-1

$$123.45 \times 23.4 = 2888.73$$

$$123.45 \times 42.6 = -5258.97$$

$$123.45 \times 51 = 6295.95$$

$$\quad \quad \quad \underline{3925.71}$$

The individual products accumulate in the memory by using the $\boxed{M}$ and $\boxed{M}$ keys.

$\boxed{M}$: Memory equals addition key

$\boxed{M}$: Memory equals subtraction key

Division by constant with accumulation

7-2

$$4578 \div 360 = 12.72$$

$$2902 \div 360 = 8.06$$

$$8716 \div 360 = 24.21$$

$$\quad \quad \quad \underline{- 3.43}$$

The individual quotients accumulate in the memory by using the $\boxed{M}$ and $\boxed{M}$ keys.

7-1

Operation	Printed tape
DP=2 $\frac{5}{4}$ $\boxed{CM}$	0 $\boxed{CM}$
123.45 $\boxed{\times}$	123.45 $\times$
23.4 $\boxed{M}$	23.4 =
	2888.73 $\boxed{M}$
42.6 $\boxed{M}$	42.6 =
	5258.97 $\boxed{M}$
51 $\boxed{M}$	51 =
	6295.95 $\boxed{M}$
	3925.71 $\boxed{CM}$
DP=2 $\frac{5}{4}$ $\boxed{CM}$	0 $\boxed{CM}$
4578 $\boxed{\div}$	4578 $\div$
360 $\boxed{M}$	360 =
	12.72 $\boxed{M}$
2902 $\boxed{M}$	2902 =
	8.06 $\boxed{M}$
8716 $\boxed{M}$	8716 =
	24.21 $\boxed{M}$
	3.43 $\boxed{CM}$

red

8. DIVIDE PRORATION

8-1

$$\begin{array}{r}
 123456789 \\
 123+456+789 \\
 \hline
 123456789 \\
 123+456+789 \\
 \hline
 123456789 \\
 123+456+789 \\
 \hline
 \end{array}
 \begin{array}{l}
 \times 123 = 11100281 \\
 \times 456 = 41152263 \\
 \times 789 = 71204245 \\
 \hline
 123456789
 \end{array}$$

※ This intermediate quotient is the constant multiplicand. Thereafter, simply enter the variable multipliers and depress the $\boxed{M}$ key.

Pressing the $\boxed{M}$ key is used to compare the sum with the initial value which is divided proportionately.

8-1

Operation	Printed tape
DP=0 $\boxed{54}$	
$\boxed{CM}$	0 CM
$\boxed{C}$	0 C
123456789 $\boxed{+}$	123456789 $\div$
123 $\boxed{+}$	123 +
456 $\boxed{+}$	456 +
789 $\boxed{+}$	789 +
$\boxed{=}$	1368 *
※ $\boxed{\times}$	1368 $\times$
123 $\boxed{M}$	123 =
	11100281 $\overset{M}{+}$
456 $\boxed{M}$	456 =
	41152263 $\nearrow \overset{M}{+}$
789 $\boxed{M}$	789 =
	71204245 $\nearrow \overset{M}{+}$
$\boxed{CM}$	123456789 CM

9. INVOICE CALCULATION

9-1

Quantity	Unit Price	Price
11	1.23	13.53
12	4.11	49.32
3	2.03	6.09
		<u>68.94</u> ◇
Discount 10%		6.89 -
		62.05 ◇
Sales Tax 5%		3.10 +
Cost of Transportation		2.50 +
		<u>67.65</u> ×

9-1

Operation	Printed tape
DP=2 54	
11 ☐ CM	0 CM
12 ☒ X	11 X
1.23 ☐ W	1.23 =
	13.53 M+
12 ☒ X	12 X
4.11 ☐ W	4.11 =
	49.32 M+
3 ☒ X	3 X
2.03 ☐ W	2.03 =
	6.09 M+
	68.94 RM
☐ X	68.94 X
10 ☐ %	10 %
	6.89 ×
☐ M	6.89 M
☐ RM	62.05 RM
☒ X	62.05 X
5 ☐ %	5 %
	3.10 ×
☐ W	3.10 M+
2.50 ☐ W	2.50 M+
☐ CM	67.65 CM

10. APPLICATION OF MEMORY AND ACCUMULATOR

The amount sold and average price.

10-1	Quantity	Unit Price	Price
	10	2.38	23.80
	20	1.38	27.60
	15	3.65	54.75
	45		106.15

Average price: $\frac{106.15}{45} = 2.35$

10-1

Operation	Printed tape
DP=2 $\overline{N}$	
$\boxed{CM}$	0 CM
$\boxed{C}$	0 C
10 $\boxed{M}$	10.00 M
$\boxed{\times}$	10.00 X
2.38 $\boxed{=}$	2.38 =
	23.80 X
$\boxed{+}$	23.80 +
20 $\boxed{M}$	20.00 M
$\boxed{\times}$	20.00 X
1.38 $\boxed{=}$	1.38 =
	27.60 X
$\boxed{+}$	27.60 +
15 $\boxed{M}$	15.00 M
$\boxed{\times}$	15.00 X
3.65 $\boxed{=}$	3.65 =
	54.75 X
$\boxed{+}$	54.75 +
$\boxed{\div}$	106.15 :
$\boxed{CM}$	45.00 CM
$\boxed{=}$	45.00 =
	2.35 X

11. SQUARE ROOT Note: SQUARE ROOT MODEL ONLY

11 1 $\sqrt{152417543.0625} = 12345.73$

STANDARD DEVIATION

11-2 DP=6 $\sqrt{4}$

$$\sigma = \sqrt{\frac{n(\sum x^2) - (\sum x)^2}{n^2}}$$

VALUES OF x : 2. 3. 4. 5. 6

$n = 5$

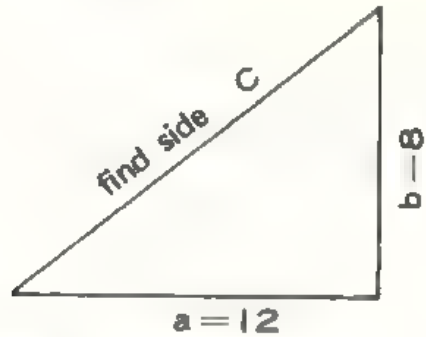
$\sigma = 1.414214$

Operation	Printed tape
DP=2 FL	
152417543.0625 $\square$	152417543.0625 $\sqrt{}$
	12345.73 $\times$
DP=6 $\sqrt{4}$	
2 $\square$	2.000000 M
$\times$	2.000000 $+$
$=$	2.000000 $\times$
	4.000000 $=$
	4.000000 $\times$
$+$	4.000000 $+$
3 $\square$	3.000000 M
$\times$	3.000000 $+$
$=$	3.000000 $\times$
	3.000000 $=$
	9.000000 $\times$
$+$	9.000000 $+$
4 $\square$	4.000000 M
$\times$	4.000000 $+$
$=$	4.000000 $\times$
	4.000000 $=$
	16.000000 $\times$
$+$	16.000000 $+$
5 $\square$	5.000000 M
$\times$	5.000000 $+$
$=$	5.000000 $\times$
	5.000000 $=$
	25.000000 $\times$

Operation		Printed tape	
	$\boxed{+}$	25.000000	+
6	$\boxed{M}$	6.000000	$\overset{+}{M}$
	$\boxed{\times}$	6.000000	$\times$
	$\boxed{=}$	6.000000	$=$
		36.000000	$\times$
	$\boxed{+}$	36.000000	+
	$\boxed{\times}$	90.000000	$\times$
5	$\boxed{=}$	5	$=$
		450.000000	$\times$
	$\boxed{+}$	450.000000	+
	$\boxed{CM}$	20.000000	$\overset{C}{M}$
	$\boxed{\times}$	20.000000	$\times$
	$\boxed{-}$	20.000000	-
		400.000000	$\times$
	$\boxed{-}$	400.000000	-
	$\boxed{\div}$	50.000000	:
5	$\boxed{=}$	5	$=$
		10.000000	$\times$
	$\boxed{=}$	10.000000	$=$
		2.000000	$\times$
	$\boxed{\sqrt{\quad}}$	2.000000	$\sqrt{\quad}$
		1.414214	$\times$

PYTHAGOREAN THEOREM

11-3



Hypotenuse
(Side C) = $\sqrt{\text{Side } a^2 + \text{Side } b^2}$

$$C = \sqrt{a^2 + b^2}$$

$$C = \sqrt{208}$$

$$C = 14.422205$$

11-3

Operation	Printed tape
DP=6 $\frac{5}{4}$	
$\boxed{C}$	0 C
$\boxed{CM}$	0 CM
12 $\boxed{\times}$	12 $\times$
$\boxed{M}$	144.000000 M
	+
8 $\boxed{\times}$	8 $\times$
$\boxed{M}$	8 =
	64.000000 M
$\boxed{CM}$	+
$\boxed{\sqrt{\quad}}$	208.000000 CM
	208.000000 $\sqrt{\quad}$
	14.422205 $\times$

12. CAPACITY OF INPUT BUFFER

The UNICOM 141 contains a powerful 8 word input buffer. It scans the keyboard 40 times per second regardless of calculating or printing.

Therefore, the information on the keyboard, automatically, can be read into the input buffer. After completion of the preceding calculation, the function read in the input buffer, sequentially, will be executed.

13. CAPACITY OF NUMBER ENTRY

The number entry capacity of the UNICOM 141 electronic calculator is 14 digits, plus decimal point and sign.

14. CAPACITY IN ADDITION/SUBTRACTION

The capacity of accumulators and memory is 14 digits, plus decimal point and sign.

14-1 DP=6 $\sqrt{N}$
12345678 $\oplus$

14-2 DP=6 $\sqrt{N}$
123456789 $\oplus$

Before addition (also subtraction), the contents of entry register, automatically, is adjusted to the decimal places designated by D.P. Selector.

14-3 DP=8 $\sqrt{N}$
 $900000 + 100000 = 1000000.00000000$
(15 digits overflow)

The contents of both accumulators (also memory), automatically, can be set to the old figures.

To get the previous figure, depress the $\odot$ key, the $\oplus$ key and the $\ominus$ key in regular sequence.

14-1

14-2

14-3

Operation	Printed tape
DP=6 $\sqrt{N}$ 12345678 $\oplus$	12345678.000000 +
DP=6 $\sqrt{N}$ 123456789 $\oplus$	
DP=8 $\sqrt{N}$ 900000 $\oplus$ 100000 $\oplus$	900000.00000000 + 100000.00000000 +
$\odot$ $\oplus$	0.00000000 +
$\ominus$	900000.00000000 $\times$

15. CAPACITY IN MULTIPLICATION

15-1 FL

$$123456789 \times 100000 = 12345678900000$$

15-2 FL

$$123456789 \times 1000000 = 123456789000000$$

(15 digits overflow)

In FLOATING mode (also intermediate product in FIXED mode), the number of integers in the product of a multiplication cannot exceed 14.

The constant multiplicand is still retained in the multiplication-division register.

15-3 DP=8 $\overline{54}$

$$123456 \times 1 = 123456.00000000$$

(14 digits)

15-4 DP=8 $\overline{54}$

$$1234567 \times 1 = 1234567.00000000$$

(15 digits overflow)

In FIXED mode, the number of integers in the product of a multiplication with the $\boxed{=}$ key cannot exceed the difference of the DP setting and 14.

15-1

15-2

15-3

15-4

Operation	Printed tape
FL	
123456789 $\boxed{\times}$	123456789 $\times$
100000 $\boxed{=}$	100000 $=$
	12345678900000 $\times$
FL	
123456789 $\boxed{\times}$	123456789 $\times$
1000000 $\boxed{=}$	1000000 $=$
$\times$	
DP=8 $\overline{54}$	
123456 $\boxed{\times}$	123456 $\times$
1 $\boxed{=}$	1 $=$
	123456.00000000 $\times$
DP=8 $\overline{54}$	
1234567 $\boxed{\times}$	1234567 $\times$
1 $\boxed{=}$	1 $=$
	

16. CAPACITY IN DIVISION

16-1 FL

$$4000000 \div 0.0000003 = 13333333333333$$

(14 digits)

16-2 FL

$$40000000 \div 0.0000003 = 133333333333333$$

(15 digits overflow)

In FLOATING mode (also intermediate quotient), the number of integers in the quotient of division cannot exceed 14.

*The constant divisor is still retained in the multiplication-division register.

16-3 DP=8 $\overline{N}$

$$400000 \div 3 = 133333.33333333$$

(14 digits)

16-4 DP=8 $\overline{N}$

$$4000000 \div 3 = 1333333.33333333$$

(15 digits overflow)

In FIXED mode, the number of integers in the quotient of division with the $\boxed{=}$ key cannot exceed the difference of the DP setting and 14.

16-1

16-2

16-3

16-4

Operation	Printed tape
FL 4000000 $\boxed{\div}$ 0.0000003 $\boxed{=}$	4000000 $\div$ 0.0000003 $=$ 13333333333333 $\times$
FL 40000000 $\boxed{\div}$ 0.0000003 $\boxed{=}$	40000000 $\div$ 0.0000003 $=$
DP=8 $\overline{N}$ 400000 $\boxed{\div}$ 3 $\boxed{=}$	400000 $\div$ 3 $=$ 133333.33333333 $\times$
DP=8 $\overline{N}$ 4000000 $\boxed{\div}$ 3 $\boxed{=}$	4000000 $\div$ 3 $=$

MEMO

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